

Microcontrollers

Leading the Way in Innovations

Intelligence and power efficiency are the key driving features of devices in which microcontrollers are used. India is one of the potential markets for microcontrollers, with high-volume requirement for application-specific microcontrollers

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Demand for intelligent products and applications has given a big boost to the use of microcontrollers even in the smallest and the simplest application. Affordable pricing and high integration are catalysing the growth of microcontrollers in a variety of applications. Microcontrollers in India have seen consistent growth over last few years and hold great potential to fit inside the applications with newer functionalities.

India is one of the strategic markets for many MCU (microcontroller unit) vendors, especially for application-specific products. Some of the leading vendors in the Indian market include Atmel, Freescale, Fujitsu,

Infineon, Microchip, NXP Semiconductors, Renesas Electronics, STMicroelectronics and Texas Instruments.

Market facts and figures

Databeans expects the global microcontroller market to reach \$12.3 billion in 2010, up 14 per cent from 2009, and further to \$18.5 billion in 2015. The MCU market continues to be driven primarily by the automotive and industrial segments, although communication application designs (latest cellphones and smartphones) too are now contributing significantly to the growth, according to Databeans.

According to Technavio Insights' research, the Asia Pacific region will witness the highest growth rate during the period 2009 to 2010. The report says

that major drivers for the growth are smaller and lighter products like mobile phones and other portable gadgets, smart metering, LED lighting and touch-screen applications market. Growing preference for environment-friendly products like hybrid and electrical cars is also boosting this market.

A report by market research firm Semicast Research shows that automotive and industrial (including medical) applications are expected to be the two key growth sectors, with

total revenues for ARM-based microcontrollers and embedded microprocessors set to increase at a CAGR of 21 per cent over the period 2007-13.

Key applications

The demand for microcontrollers has increased across most of the application segments. Consumer electronics, automotive and industrial manufacturers are focused on incorporating more features into products while reducing the total power consumption. Enhancements in the MCU technology and higher peripheral integration have also led to many new applications in medical, lighting, and sensor-based industrial and home automation.

Energy. Microcontrollers are being used for smart energy management. Speaking of alternative energy systems, MCUs are being used in such applications as solar energy products (photovoltaic systems and DC-AC power inverters), wind turbines and heating systems. The world's leading economies are looking at smart grids that address the challenge of efficient energy management and distribution, and record consumer usage in great detail. This enables greater control and power savings. Both LED street lightings and solar lightings need efficient usage of the existing power, which can be controlled through advanced microcontroller architectures.

Automotive electronics. Microcontrollers are increasingly being used in automobiles for infotainment, engine

